Cyber-Attack Detection in Noisy Network Traffic

A Multi-Class Machine Learning Approach for Intrusion Detection

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Executive Summary

Objective

Develop robust classification system to detect diverse cyber-attack types within highly noisy network flow data

The Challenge

1.7M+ rows with label noise, extreme outliers, and massive class imbalance (1,588 unique labels)

Key Results

78% overall classification accuracy using gradient-boosted decision trees

Dataset Profile

Raw Data

combined_noisy.csv

Volume

1,775,435 rows × 84 columns

Data Types

- 64 float64 features
- 6 int64 features
- 14 categorical (object)

Critical Network Features

Flow Duration

Time span of network connections

Total Fwd Packets

Forward packet counts in flow

Src/Dst Ports

Source and destination ports

Inter-Arrival Times

Time between packet arrivals

Data Quality Challenges

Label Noise

1,588 unique labels with typos: "BENIGN" vs. "BNIGN", "PcortScan" vs. "PortScan"

Physical Anomalies

Negative values in non-negative features like Src Port and Flow Duration

Extreme Outliers

26% outlier rate in Idle times and packet lengths

Feature Engineering Process

Dimensionality Reduction

Dropped Flow ID, Src IP, Dst IP, Timestamp

Data Cleaning

Removed 7,304 duplicates, replaced infinities with NaN

Normalization

RobustScaler for outlier-resistant standardization

Encoding

LabelEncoder for 1,500+ noisy categories

Modeling Strategy

Algorithm

LightGBM / Gradient Boosting

Validation

Stratified K-Fold across 1,588 labels

Hardware

NVIDIA Tesla T4 GPU acceleration

Loss Function

Multi-class classification optimization

Performance Evaluation

86%

Overall Accuracy

Across 1,588 attack classes

98%

Majority Classes

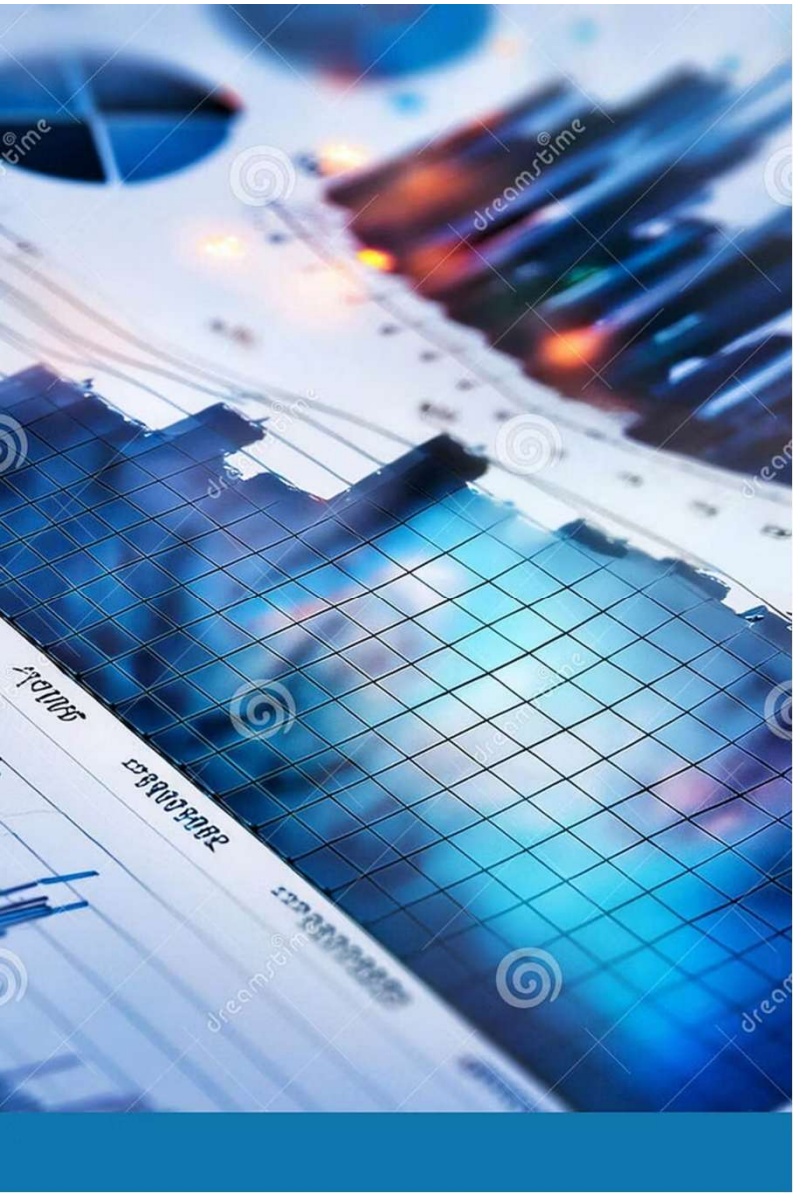
BENIGN and DoS Hulk F1-scores

0%

Noisy Classes

Typo-ridden labels like "PortScban"

Extreme class imbalance visualized using log-scale distribution



Key Insights

Noise Bottleneck

Label noise primary limitation; merging variants into parent classes would boost performance

Feature Importance

Temporal features (Idle Mean, Flow Duration) and packet sizes strongest indicators

Future Recommendations

01

Apply SMOTE or class-weighting

Improve detection of rare attacks

02

Integrate text-similarity filter

Automatically fix label typos

03

Standardize parent classes

Merge all DoS variants and similar attack types